



Semi-supervised heterogeneous graph contrastive learning for drug–target interaction prediction

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ABSTRACT

Identification of drug–target interactions (DTIs) is an important step in drug discovery and drug repositioning. In recent years, graph-based methods have attracted great attention and show advantages on predicting potential DTIs. However, these methods face the problem that the known DTIs are very limited and expensive to obtain, which decreases the generalization ability of the methods. Self-supervised contrastive learning is independent of labeled DTIs, which can mitigate the impact of the problem. Therefore, we propose a framework SHGCL-DTI for predicting DTIs, which supplements the classical semi-supervised DTI prediction task with an auxiliary graph contrastive learning module. Specifically, we generate representations for the nodes through the neighbor view and meta-path view, and define positive and negative pairs to maximize the similarity between positive pairs from different views. Subsequently, SHGCL-DTI reconstructs the original heterogeneous network to predict the potential DTIs. The experiments on the public dataset show that SHGCL-DTI has significant improvement in different scenarios, compared with existing state-of-the-art methods. We also demonstrate that the contrastive learning module improves the prediction performance and generalization ability of SHGCL-DTI through ablation study. In addition, we have found several novel predicted DTIs supported by the biological literature. The data and source code are available at: <https://github.com/TOJSSE-iData/SHGCL-DTI>.

1. Introduction

Drug–target interaction prediction is an important step in drug discovery and drug repositioning, aiming to identify the targets for drugs and medicines [1]. However, identifying potential drug–target interactions (DTIs) by biological experimental methods is very time and cost-consuming [2]. Therefore, many researchers have tried to apply computational methods to provide the potential DTIs for the subsequent biological experimental validation.

The traditional computational methods can be divided into three types: docking-based methods [3,4], ligand-based methods [5,6], and machine learning based methods [1,7–10]. The performance of docking-based methods is limited when the 3D structure of the protein is unknown [11]. Also the performance of ligand-based methods is limited when the target has only a small number of known ligands [12].

Machine learning-based methods convert DTI prediction into a classification task to predict the presence or absence of a DTI. The key assumption behind these methods is that similar drugs may share similar targets and vice versa. [1] NetLapRLS [7] applies Laplace regularized the least square to predict DTIs with known DTIs, chemical structure as well as genomic sequence data. DTINet [1] learns low-dimensional representations of drugs and proteins from multiple networks and applies inductive matrix completion to predict DTIs. These machine learning-based methods usually focus more on the first-order connections between nodes instead of high-order connections which is proved to be useful in related domains [13,14].

In the field of DTI prediction, in addition to the widely used traditional methods, methods based on graph have attracted a lot of attention, which can be considered as valuable and essential additions to the computational tool box. Methods based on graph neural network

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(GNN) have been applied to DTI prediction [15–19] in order to learn the rich topological information and the complex interactions of drugs and proteins in biological networks. Such methods usually consist of three steps: first to construct the graph with drug and protein related data, second to extract the node representations from the graph by GNN and third to use the node representations to predict DTIs. DTI-GAT [16] applies molecular fingerprints as drug features, obtains the features of proteins through pseudo position specific scoring matrices (PsePSSM), and applies the graph attention network to extract node representations. NeoDTI [17] and EEG-DTI [18] both integrate information from the heterogeneous network and learn node representations of drugs and targets by applying graph convolutional network (GCN) to heterogeneous networks. NeoDTI predicts DTIs by reconstructing the original heterogeneous network, and EEG-DTI applies the inner product method to predict DTIs. In addition to these DTI prediction methods, methods such as GraphDTA [20] and HGRL-DTA [21] would extract node features from the molecular graph to predict drug–target binding affinity.

However, methods based on graph neural networks often face the problem that the known DTIs are very limited and expensive to obtain. Recently, self-supervised contrastive learning has shown great potential in many graph representation tasks with no label. By maximizing the similarity between positive pairs and minimizing the similarity between negative pairs, contrastive learning is able to learn meaningful representations. There are several papers using graph contrastive learning methods that have achieved satisfactory results on node classification tasks [22–25] and recommendation tasks [26–28]. In addition, some studies in biology [29–31] have applied graph contrastive learning to DDI and DTI prediction tasks. However, they do not take full advantage of the rich semantic information in heterogeneous graphs and the complex interactions of heterogeneous data.

To solve the problem that the known DTIs are very limited and expensive to obtain, we propose a DTI prediction framework called Semi-supervised Heterogeneous Graph Contrastive Learning for Drug–Target Interaction prediction (SHGCL-DTI) to mitigate the impact of the problem and improve the DTI prediction performance and generalization ability. SHGCL-DTI constructs a neighbor view encoder and a meta-path view encoder for node aggregation and obtains node representations by graph contrastive learning to improve the performance of DTI prediction and the generalization ability of the framework. Specifically, SHGCL-DTI firstly constructs a heterogeneous graph with 4 types of nodes and 8 types of edges, and then initializes the features for 4 types of nodes. We take the molecular fingerprints as drug features and take the relative ratios of amino acids and dipeptides as protein features. We apply principal component analysis (PCA) to descend the dimension of drug and protein features. The features of disease and side effect are initialized by Node2vec [32]. After that, we extract node features from the neighbor view and meta-path view by a Heterogeneous Graph Convolutional Network (HGCN) and a GCN with a semantic attention mechanism respectively. We perform a graph contrastive learning task between the two views. Finally, we apply the DistMult [33] to reconstruct the original heterogeneous graph to obtain the results of DTI prediction. On the public dataset, we demonstrate that SHGCL-DTI outperforms other state-of-the-art methods in the cases of unbalanced labels, less redundant DTIs, no priori knowledge and masking partial known DTIs. Ablation study shows that contrastive learning module can improve the generalization ability and the prediction performance. Initialization strategy analysis indicates that our initialization strategy is effective in DTI prediction. Sensitivity analysis shows more details of the framework. Prediction result verification with literature support proves that SHGCL-DTI can provide reliable potential DTIs. Overall, SHGCL-DTI is expected to be a satisfactory framework for DTI prediction.

2. Materials and methods

2.1. Dataset

In order to evaluate the performance of SHGCL-DTI on DTI prediction task, we run experiments with SHGCL-DTI on Luo's dataset [1].

There are four types of nodes in the dataset, including drug, protein, disease and side-effect. Drug nodes are extracted from Drugbank [34], protein nodes are extracted from HPRD database [35], disease nodes are extracted from Comparative Toxicogenomics Database [36] and side-effect nodes are extracted from SIDER database [37].

Correspondingly, there are eight types of edges in the dataset, including six interactions or associations and two similarities. Drug–drug interaction network and drug–protein interaction network are extracted from Drugbank [34]. The drug–protein interaction is also called known DTI. Drug–disease association network and protein–disease association network are extracted from Comparative Toxicogenomics Database [36]. Protein–protein interaction network is extracted from HPRD database [35]. Drug–side effect association network is extracted from SIDER [37]. In addition to the six edges, there are also two kinds of edges based on similarity: drug–drug similarity calculated based on the chemical structure of the drug by the Tanimoto coefficient [38] and protein–protein similarity calculated based on genomic sequences by Smith–Waterman algorithm [39]. Interactions and associations are represented as edges. Drug or protein similarity means that every drug/protein is connected to each other by an edge, respectively. The same two nodes can be connected by more than one edge, e.g. two drugs can be linked by a *drug – drug – interaction* edge and a *drug – similarity* edge simultaneously. For these six interaction or association networks, the weight q of an edge is 1. For two similar networks, the weights q of the edges are determined by the similarity score between drugs [38] or the similarity score between proteins [39], and we normalize the similarity score to the range of 0 and 1 as the weights of the edges.

The nodes and edges can compose a biological network and the biological network can be represented by a heterogeneous network. A heterogeneous network can be defined as a graph $G = (\mathcal{V}, \mathcal{E})$. $\mathcal{V}$ and $\mathcal{E}$ denote the set of nodes and the set of edges respectively. And let $\mathcal{A}$ and $\mathcal{R}$ denote the set of node types and edge types respectively and there are $\mathcal{A} = \{\text{drug, protein, disease, side-effect}\}$ and $\mathcal{R} = \{\text{drug–drug interaction, drug–protein interaction, drug–disease association, drug–side-effect association, protein–protein interaction, protein–disease association, drug–drug similarity, protein–protein similarity}\}$. More details are shown in Supplementary Tables S1 and S2.

2.2. Node feature initialization

The first step is the initialization of the drug nodes. Although Simplified Molecular Input Line Entry Specification (SMILES) is often adapted to extract the chemical structure of drugs [40,41], subtle differences in functional groups can lead to significant differences in the chemical properties of drugs, even if their SMILES codes are similar [15]. Also, in some studies [42,43], it is shown that the descriptors of molecular structure fingerprints can effectively represent the drug features. For drug nodes, we obtain MACCS fingerprints based on SMILES expression [44], which defines 166-bit molecular descriptors. In order to reduce the unimportant molecular descriptors, we apply Principal Components Analysis (PCA) with 128 principal components to descend the dimension of the molecular descriptors of the drug nodes as input.

The second step is to initialize the feature of the protein nodes. Previous study [15] also shows that the chemical properties and relative ratio of amino acids in proteins can represent the feature of proteins. Therefore, for the protein nodes, we extract the relative ratio of amino acids and dipeptides from the protein sequences and construct 420 descriptors. As the same as dealing with the drugs, we applied PCA with 128 principal components to descend the dimension of the descriptors of the protein nodes as inputs.

For disease and side effect nodes initialization, to the best of our knowledge, there is no better solution to directly convert their own attributes into features so far. Therefore, we adapted a method called node2vec [32] to initialize and construct them, which can convert

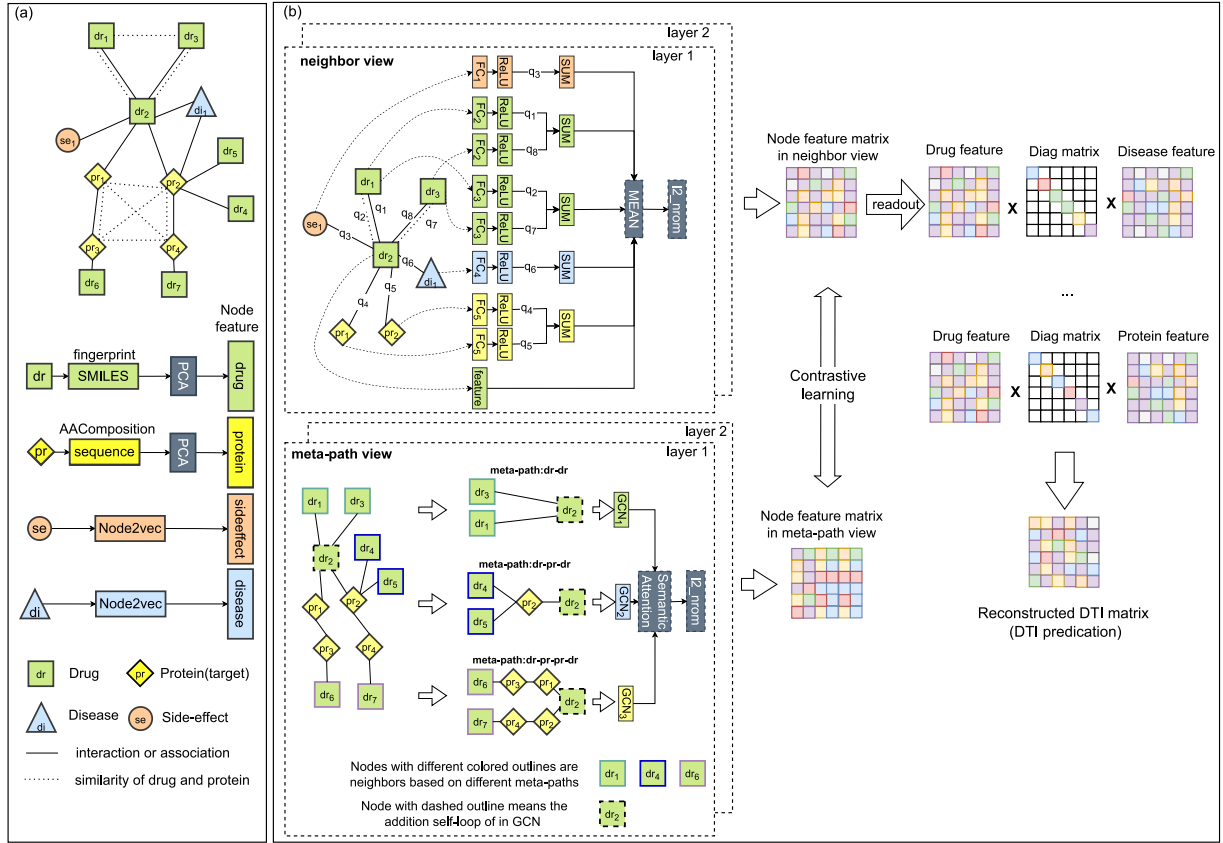


Fig. 1. The workflow of SHGCL-DTI. SHGCL-DTI is divided into two main steps. (a) In the first step, we construct a heterogeneous network and initialize the node features. In this heterogeneous network, there are four types of nodes and eight types of edges. Among them, dr represents drug, pr represents protein (target), se represents side effect, and di represents disease. The eight types of edges include drug–drug interaction, drug–side effect association, drug–disease association, drug–protein interaction, protein–protein interaction, protein–disease association, drug–drug similarity, and protein–protein similarity. For drug nodes, molecular fingerprints are extracted by using SMILES expressions and then reduce dimensionality. For protein nodes, relative ratios of amino acids and dipeptides are extracted by using protein sequences and then reduce dimensionality, while side effects and disease nodes features are initialized by node2vec. (b) We take node dr_2 as an example to describe the process of generating node representation. In the neighbor view, different types of edges are considered in the process of generating representations. $q_1, q_2, \dots, q_8$ denote the weights of the edges. In the meta-path view, different types of meta-paths are considered, and meta-path-based neighbor's information of different meta-paths is aggregated with semantic-level attention. And we obtain the representation after contrastive learning. Then, we reconstruct the original heterogeneous network using the representation of the nodes which is derived from the neighbor view to obtain the DTI prediction.

the relationships within context into features. Node2vec is performed on drug–drug interaction, drug–disease association, drug–side effect association, protein–protein interaction and protein–disease association. The output dimension is 128 which is the same as drug and protein features.

2.3. SHGCL-DTI

2.3.1. Neighbor view based graph encoder

Recently, graph convolutional networks (GCN) [45] have been widely adapted to extract features from graph-structured data, which enables the aggregations of the features for neighbor nodes. Reasonably, to extract features from heterogeneous networks constructed in the previous step, we use a heterogeneous graph convolutional network (HGCN) inspired by [46].

In a heterogeneous network, different types of edges will be aggregated respectively [46]. As shown in Fig. 1(b), in the neighbor view, for each layer of HGCN, we have the following equation:

$$h_i^{(l+1)} = \frac{1}{1 + |\mathcal{R}|} \left(\sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_i^r} \frac{q_{i,j,r}}{Q_{i,r}} \sigma(\mathcal{W}_r^{(l)} h_j^{(l)} + b_r^{(l)}) + h_i^{(l)} \right) \quad (1)$$

where l represents the number of layers, $\mathcal{N}_i^r$ represents the set of neighbors of v_i and $|\mathcal{R}|$ is the number of the edge types. $Q_{i,r} = \sum_{j \in \mathcal{N}_i^r} q_{i,j,r}$, where $q_{i,j,r}$ represents the weight of edge. $\mathcal{W}_r^{(l)}$ is the weight matrix and $b_r^{(l)}$ is a bias. $\sigma(\cdot)$ is an activation function ReLU.

$h_i^{(l)}$ and $h_j^{(l)}$ represent the target node's features and neighbors' features respectively. In particular, when $l = 0$, we apply the initialized features as inputs. In this step, we can obtain the node representations extracted from the neighbor view.

When there is only one layer of HGCN, the node can only aggregate information from first-order neighbors. Stacking N layers of graph convolution layers allows nodes to aggregate information from N -order neighbors. The larger the N is, the wider the perceived range is. When N reaches a certain level, the gradient disappears which leads to over-smoothing. When it occurs, the representation of nodes within the same connected component will converge [47]. Therefore, we implement a two-layer HGCN in this framework, with each layer extracting features from the heterogeneous network. We also find that L2 normalization helps to improve the performance of prediction. Finally, we obtain the node representation h^{ni} .

2.3.2. Meta-path view based graph encoder

A meta-path is a sequence of multiple node types connected by relation types, which can be represented by $\mathcal{P} = a_1 \xrightarrow{r_1} a_2 \xrightarrow{r_2} \dots \xrightarrow{r_n} a_{n+1}$, where $a_i \in \mathcal{A}$ and $r_i \in \mathcal{R}$. Given a node v_i and a meta-path $\mathcal{P}_n$, the meta-path based neighbors of v_i are defined as the nodes connected to v_i via meta-path $\mathcal{P}_n$. In addition, we enforce self-loop to each node, which means that the node v_i itself is also its neighbor. Specifically, we take a meta-path $drug_1 \xrightarrow{dri} protein \xrightarrow{idi} drug_2$ as example, the neighbors of $drug_1$ are $drug_1$ and $drug_2$.

In a heterogeneous network, there is rich semantic information between different nodes and edges, which cannot be captured by traditional GCNs. In order to effectively utilize the semantic information in the heterogeneous network, different meta-paths are adapted to capture this information. By the predefined meta-paths, we can obtain node representations using a graph encoder in the meta-path view. As shown in Fig. 1(b), for each meta-path in the meta-path view, we apply a meta-path-based GCN to extract features from the meta-path-based neighbors:

$$h_{i,P_n}^{(l+1)} = \sum_{j \in \mathcal{N}_i^{P_n}} c_{P_n}^{i,j} \mathcal{W}^{(l)} h_j^{(l)} \quad (2)$$

where $c_{P_n}^{i,j} = 1/\sqrt{d_{i,P_n} d_{j,P_n}}$, d_{i,P_n} and d_{j,P_n} denote the number of nodes connected to v_i and v_j via meta-paths P_n respectively. P_n denotes a kind of meta-path. And h_i and h_j represent the feature of node v_i and node v_j . $\mathcal{N}_i^{P_n}$ are the neighbors v_i based on the meta-path P_n . On m meta-paths, we can get m representations $h_{i,P_1}^l, h_{i,P_2}^l, \dots, h_{i,P_m}^l$. Because the different types of meta-paths contain different semantic information, we apply semantic-level attention when we aggregate the results of meta-paths. So, we calculate the importance of each meta-path firstly:

$$\mathcal{W}_{P_n} = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \alpha^T \cdot \sigma(\mathcal{W} h_{i,P_n}^{(l)} + b) \quad (3)$$

where $\mathcal{W}$ is the weight matrix, b is the bias vector, and α is the semantic level attention vector. After obtaining the importance of each meta-path, we normalize it by softmax to obtain the weight β_{P_n} of the meta-path P_n :

$$\beta_{P_n} = \text{softmax}(\mathcal{W}_{P_n}) = \frac{\exp(\mathcal{W}_{P_n})}{\sum_{P_i \in \mathcal{P}} \exp(\mathcal{W}_{P_i})} \quad (4)$$

The weight β_{P_n} can be interpreted as the weight of contribution by each meta-path to the feature. By computing the weights of multiple meta-paths, we can obtain the final node representations:

$$h_i^{(l)} = \sum_{n=1}^m \beta_{P_n} \cdot h_{i,P_n}^{(l)} \quad (5)$$

As the same as the neighbor view, we also apply two layers of GCN and L2 normalization in meta-path view. Finally, we obtain the node representation h^{mp} .

2.3.3. Contrastive learning

In the contrastive learning module, we define highly related node pairs, which are connected by multiple meta-paths, as positive pairs, while other node pairs are defined as negative pairs. We maximize the similarity between the representations of the two nodes in positive pairs, where the representations of the two nodes in positive pairs come from different views. Therefore, the node can learn the representation from two views. The obtained representations can be used for subsequent DTI prediction.

Specifically, we use an MLP with a hidden layer to map h^{ni} and h^{mp} to the space used to compute the contrastive loss, obtaining z^{ni} and z^{mp} .

The next step is to define the positive and negative pairs. It is a common practice in contrastive learning to directly take different representations of the same node in two views as positive pairs, and all others as negative pairs. In a graph, nodes connected by edges are related, so we can utilize these edges to determine the positive pairs. But it is difficult to choose the positive and negative pairs directly from the first-order neighbors. The meta-path provides high-order information, and the selection of positive and negative pairs from the meta-path can improve the performance of DTI prediction. Therefore, we can define that two nodes v_i and v_j , which are connected by a large number of meta-paths, can be considered positive pairs and the other nodes can be considered as negative pairs. The meta-paths are consistent between in meta-path view based graph encoder and in contrastive learning. However, the connections among different meta-paths may vary greatly. The two nodes may have a large number of

connections on just one type of meta-paths. We normalize the neighbor matrix based on meta-path to avoid that the positive pairs are heavily influenced by one type of meta-paths:

$$C_i(j) = \sum_{P_n \in \mathcal{P}} \frac{C_{i,P_n}(j)}{\sum_{u \in \mathcal{N}_i^{P_n}} C_{i,P_n}(u)} \quad (6)$$

where $C_{i,P_n}(j)$ is the number of the connections of node i and node j between meta-path P_n , and $\mathcal{N}_i^{P_n}$ is the set of neighbors of i based on the meta-path P_n . Then we define a threshold $\mathcal{T}$ and a set $S_i = \{j | j \in \mathcal{V} \text{ and } C_i(j) > 0\}$. The nodes, whose value of $C_i(j)$ are the Top $\mathcal{T}$, will be defined as positive pairs of i and compose the set of the positive samples of i called $\mathcal{Pos}_i$. The other nodes compose the set of the negative pairs, called $\mathcal{Neg}_i$. We can use $\mathcal{ALL}$ to represent $\mathcal{Pos}_i \cup \mathcal{Neg}_i$. After getting the positive and negative pairs, we apply InfoNCE to calculate the contrastive loss, where τ is the temperature coefficient.

$$\mathcal{L}_a^{ni} = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}_a} -\log \frac{\sum_{j \in \mathcal{Pos}_i} \exp(\text{sim}(z_i^{ni}, z_j^{mp})/\tau)}{\sum_{k \in \mathcal{ALL}} \exp(\text{sim}(z_i^{ni}, z_k^{mp})/\tau)} \quad (7)$$

where a is the type of nodes and $\mathcal{V}_a$ is the set of nodes with type a . And $\text{sim}(u, v)$ denotes the cosine similarity between two vectors u and v . The computation of $\mathcal{L}_a^{mp}$ is similar to that of $\mathcal{L}_a^{ni}$, except that the representation of the target node is replaced by z_i^{mp} :

$$\mathcal{L}^{cl} = \sum_{a \in \mathcal{A}} \mathcal{L}_a^{ni} + \mathcal{L}_a^{mp} \quad (8)$$

In order to better reconstruct the original heterogeneous network, we apply h_i^{ni} which aggregates the information of the first-order neighbors to predict DTIs.

2.4. DTI prediction

In this section, we will introduce how to apply our proposed framework in predicting DTIs and how to optimize the framework.

After obtaining the representation of the four types of nodes, we apply DistMult [33] to predict DTIs in order to evaluate the utilization of heterogeneous network information. Specifically, for each of the eight types of edges in the original heterogeneous network, we apply DistMult to reconstruct them, in which the reconstructed DTI network is the result of DTI prediction. As shown in Fig. 1(c), taking the drug-target interaction network as an example, the reconstructed network can be calculated as follows:

$$\mathcal{G}_{dti}^{re} = \mathcal{H}_{drug}^T \mathcal{M}_{dti} \mathcal{H}_{protein} \quad (9)$$

where, $\mathcal{H}_{drug}$ and $\mathcal{H}_{protein}$ are the matrices of node representations of drugs and proteins obtained in the previous step, respectively. $\mathcal{M}_{dti}$ is a learnable diagonal matrix adapted to reconstruct the DTI network. Similarly, for the other networks, we also apply DistMult for reconstruction. All the diagonal matrices are learnable. Then the reconstruction loss can be calculated as follows:

$$\mathcal{L}^{re} = \sum_{r \in \mathcal{R}} (\mathcal{G}_r - \mathcal{G}_r^{re})^2 = \sum_{r \in \mathcal{R}} \sum_{i,j \in \mathcal{G}_r} (q_{i,j} - q_{i,j}^{re})^2 \quad (10)$$

where, $q_{i,j}$ and $q_{i,j}^{re}$ represent the weights on the original network and the weights on the reconstructed network respectively. By minimizing the reconstruction loss, we can obtain the probability of whether there is an interaction between nodes. Finally, the optimization objective of our framework consists of three parts: the reconstruction loss, the contrastive loss and L2 regularization term:

$$\mathcal{L} = \mathcal{L}^{re} + \lambda_1 \mathcal{L}^{cl} + \lambda_2 \mathcal{L}^{L2} \quad (11)$$

where $\mathcal{L}^{L2} = \sum_w \mathcal{W}^2$ is the sum of the squares of the weights, and λ_1 and λ_2 are adjustable hyper-parameters.

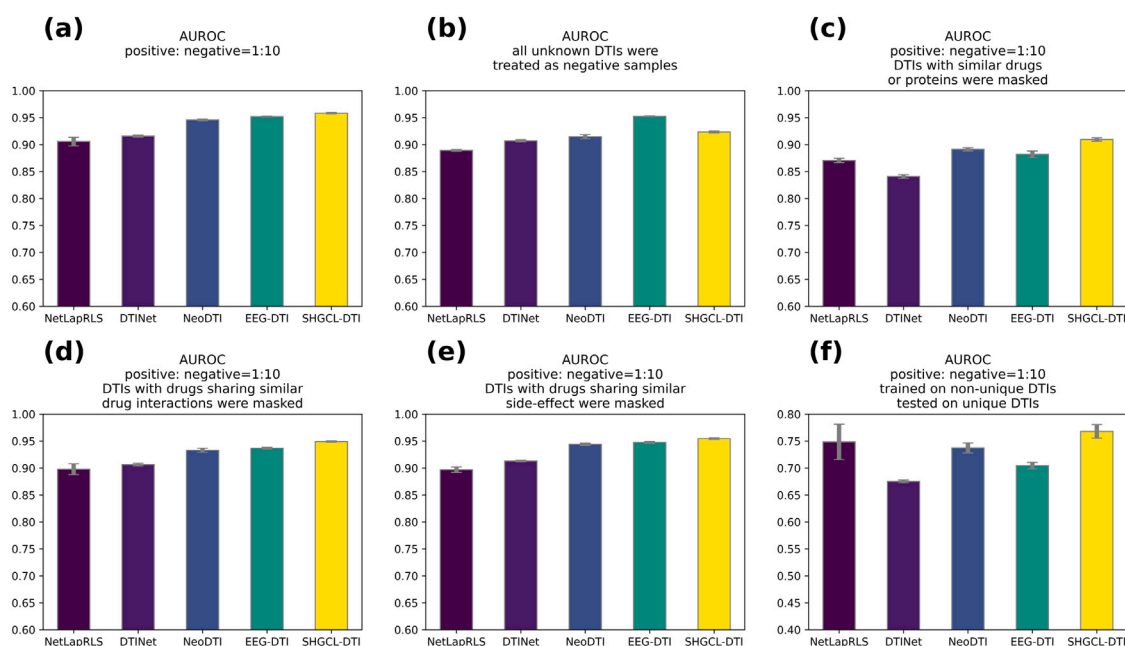


Fig. 2. The AUROC of SHGCL-DTI and SOTA baselines in different scenarios. (a) The ratio of positive and negative samples is set to 1:10. (b) All the unknown DTIs are treated as negative samples. (c) DTIs with similar drugs and proteins were masked. (d) DTIs with drugs sharing similar drug interactions were masked. (e) DTIs with drugs sharing similar side effects were masked. (f) All the non-unique DTIs were divided into the training set and unique DTIs were divided into the test set. Each experiment is repeated ten times, and the mean and standard deviation are reported.

3. Results and discussion

3.1. Baselines

We compare SHGCL-DTI with four baseline methods, two of which are advanced machine learning methods (NetLapRLS [7], DTINet [1]) and the others are advanced GNN-based methods (NeoDTI [17], EEG-DTI [18]):

NetLapRLS [7]. NetLapRLS is a semi-supervised learning method based on Laplacian regularized least square [48]. In order to improve the performance for DTI prediction, NetLapRLS considers the drug–protein interaction network as well as the chemical structure and genomic sequence data. NetLapRLS applies two classifiers trained on the drug and protein domains to give the final prediction.

DTINet [1]. DTINet is a computational pipeline to predict novel drug–target interactions (DTIs) from a heterogeneous network. DTINet applies the network diffusion algorithm (random walk with restart) and dimensionality reduction scheme (diffusion component analysis) to get low-dimensional vector representations of nodes from the heterogeneous network. Then, DTINet applies the projected low-dimensional representations to predict DTIs.

NeoDTI [17]. NeoDTI is a framework that can integrate diverse information from a heterogeneous network to predict new DTIs. Through information passing and aggregation operations, NeoDTI extracts the complex hidden features of all kinds of nodes. Then, NeoDTI reconstructs the original heterogeneous network to predict DTIs in an end-to-end manner.

EEG-DTI [18]. EEG-DTI is an end-to-end HGCN-based algorithm to predict DTIs. Through concatenating the representations in different layers, EEG-DTI learns the low-dimensional representations of drugs and targets to predict DTIs.

3.2. Experimental settings

Because the DTI prediction is a binary classification task, we chose Area Under the Receiver Operating Characteristic Curve (AUROC) and Area Under the Precision-Recall Curve (AUPR) as evaluation metrics.

The positive samples for DTI prediction are selected from the known DTIs and the negative samples are selected from the unknown DTIs.

We evaluate SHGCL-DTI and baseline methods on the same dataset with the following settings. For SHGCL-DTI, we set the number of GCN layers to 2 and the dimension of the hidden layer to 2048. We set the learning rate to $1e-3$, the temperature parameter τ to 0.5, and the dropout rate to 0.2. We run 5000 iterations using the Adam optimizer with the early stopping mechanism.

In terms of meta-path selection, as shown in Supplementary Table S3, for meta-path like drug–disease–drug, the number of edges based on drug–disease–drug is 494,028 and the number of full connections between the drug nodes is 501,264, resulting in the proportion to be 98.56%. If we apply the meta-path like drug–disease–drug or drug–side effect–drug, one drug node will aggregate almost all (98.56% and 93.11% respectively) of the other drug nodes, which makes all the drug nodes have the similar representation and results in a meaningless aggregation. The situation like this occurs when the meta-path contains disease node or side effect node. Therefore, we only choose the meta-path whose adjacency matrix based on this meta-path is sparse. In addition, since our task aims to predict DTIs, our meta-path selection strategy focuses on drugs and proteins in order to include more DTI-related information. Based on these considerations, we do not apply contrastive learning for disease and side effect nodes, because the meta-paths which contain disease and side-effect have a high proportion of the number of connected edges and full connections. Excluding these meta-paths can make the calculations faster without reducing the prediction performance. Therefore, our meta-paths are set as: drug–drug, drug–protein–drug, drug–protein–protein–drug, protein–protein, protein–drug–protein, protein–drug–drug–protein. We also set the thresholds for the number of positive samples for drug and protein to 10 and 5, respectively, which is the best choice of parameters obtained through experiments.

3.3. Experiment results

3.3.1. Comparison with baselines

In order to evaluate the effectiveness of the SHGCL-DTI, we compare SHGCL-DTI with the four advanced methods in seven different

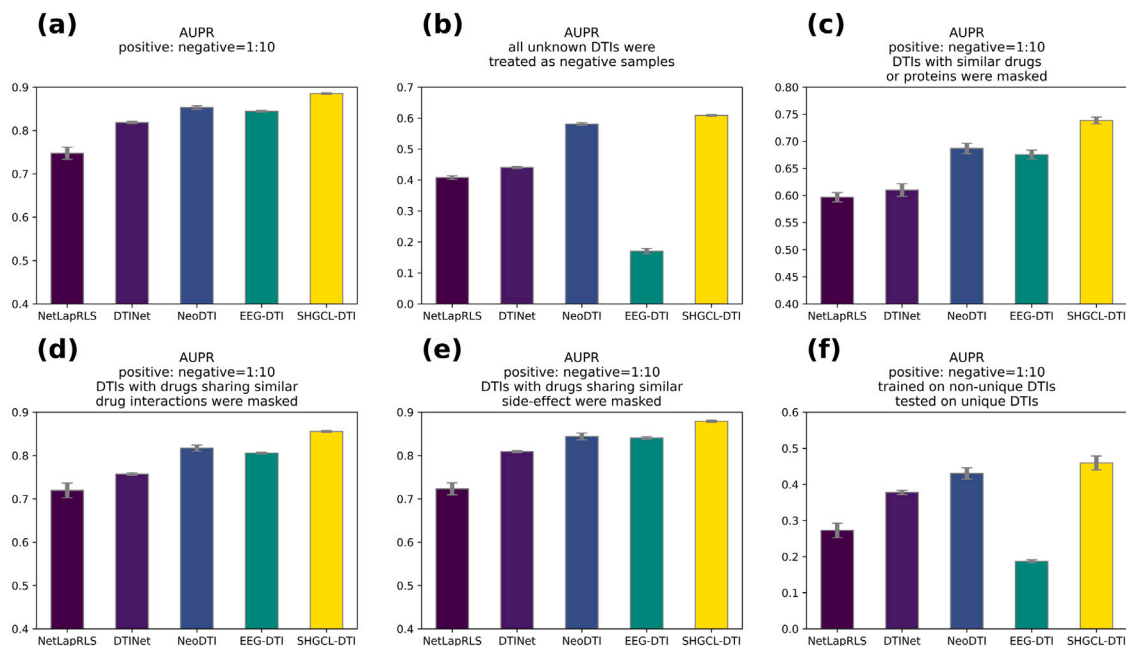


Fig. 3. The AUPR of SHGCL-DTI and SOTA baselines in different scenarios. Settings of (a)–(f) are the same as Fig. 2. Each experiment is repeated ten times, and the mean and standard deviation are reported.

scenarios. We divided the seven scenarios into four groups: scenario (a–b), scenario (c–e), scenario (f) and scenario (g), so that we can prove the improvement of SHGCL-DTI on the dataset with imbalanced labels, on the dataset with less redundant DTIs, on the dataset without priori knowledge and on the dataset with masking partial known DTIs. The experiment results are presented in Figs. 2–4.

In the first two scenarios, we choose the different ratios of positive and negative samples. In the first scenario (a), we build a dataset where the positive samples are all known DTIs and the negative samples are randomly selected. The number of negative samples is 10 times as many as the positive samples. We run a 10-fold cross-validation test for evaluation and present the results in Fig. 2(a) and 3(a). SHGCL-DTI performs better than all baseline methods, with 0.66% and 3.24% higher than the second-best results (EEG-DTI and NeoDTI, respectively) on AUROC and AUPR. Then, in the second scenario (b), we build a dataset where the positive samples are all known DTIs and the negative samples are all unknown DTIs, which makes the number of negative samples 8103 times as many as the number of positive samples. We run a 10-fold cross-validation test for evaluation and present the results in Fig. 2(b) and 3(b). SHGCL-DTI performs better than all baseline methods on AUPR, with 2.80% higher than the second best result (NeoDTI). SHGCL-DTI achieves the second best result on AUROC, which is 2.90% lower than EEG-DTI. Since AUPR is a more convincing metric on the extremely imbalanced dataset [49], we believe that SHGCL-DTI outperforms all baseline methods.

The dataset contains redundant DTIs (i.e. one protein is connected to more than one similar drugs and vice versa.), and these redundant DTIs can be easily predicted. Masking these DTIs can reduce the potential redundancy in DTIs so that we can get a more realistic and valuable result instead of an inflated result [1]. In the next three scenarios (c–e), we construct three different datasets following Luo [1]. Specifically, in the third scenario (c), the DTIs with similar drugs (drug chemical structure similarities > 0.6) and similar proteins (protein sequence similarities > 0.4) are masked. In the fourth scenario (d), DTIs with drugs sharing similar drug interactions (Jaccard similarities > 0.6) are masked. In the fifth scenario (e), DTIs with drugs sharing similar side effects (Jaccard similarities > 0.6) are masked. The three datasets are built with all known DTIs partly masked, and the negative samples are randomly selected. In each dataset, the number of negative samples

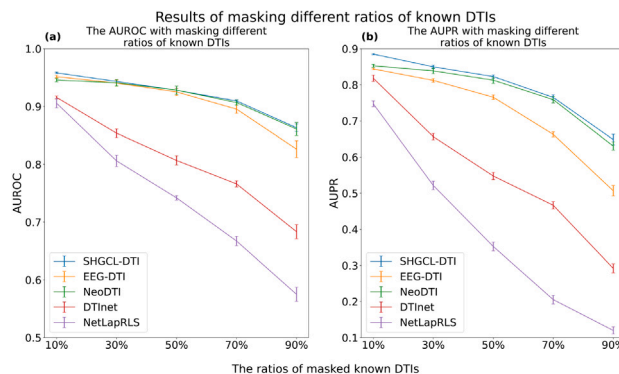


Fig. 4. Results of masking different ratios of known DTIs.

is 10 times as many as the number of positive samples. We run a 10-fold cross-validation test for evaluation and present the results in Fig. 2(c–e) and 3(c–e). As shown in Fig. 2(c–e) and 3(c–e), SHGCL-DTI still achieves the best performance on both AUROC and AUPR, which proves the stability of SHGCL-DTI.

In our dataset, there are some DTIs that the drug only has one protein and vice versa. In this scenario, the 10-fold cross-validation may not get the precise result. Therefore, In the scenario (f), in order to validate the performance of our framework in this case, we perform no priori knowledge experiments. Specifically, if a drug only interacts with one protein and vice versa, we call these DTIs unique DTIs. We take unique DTIs as the testing set. The other DTIs, called non-unique DTIs, compose the training set. The number of negative samples is 10 times as many as the number of positive samples in both the training set and testing set. By dividing the dataset in this way, the testing set is independent of the training set and the drugs and proteins are unseen in training set. The results in Fig. 2(f) and 3(f) show that SHGCL-DTI outperforms the other baseline methods on AUROC and AUPR with 1.94% and 2.91% higher than the second best result.

In scenario (g), we randomly mask known DTIs by different ratios, which allows us to demonstrate the ability of SHGCL-DTI and other methods to predict unknown DTI. Specifically, we randomly mask

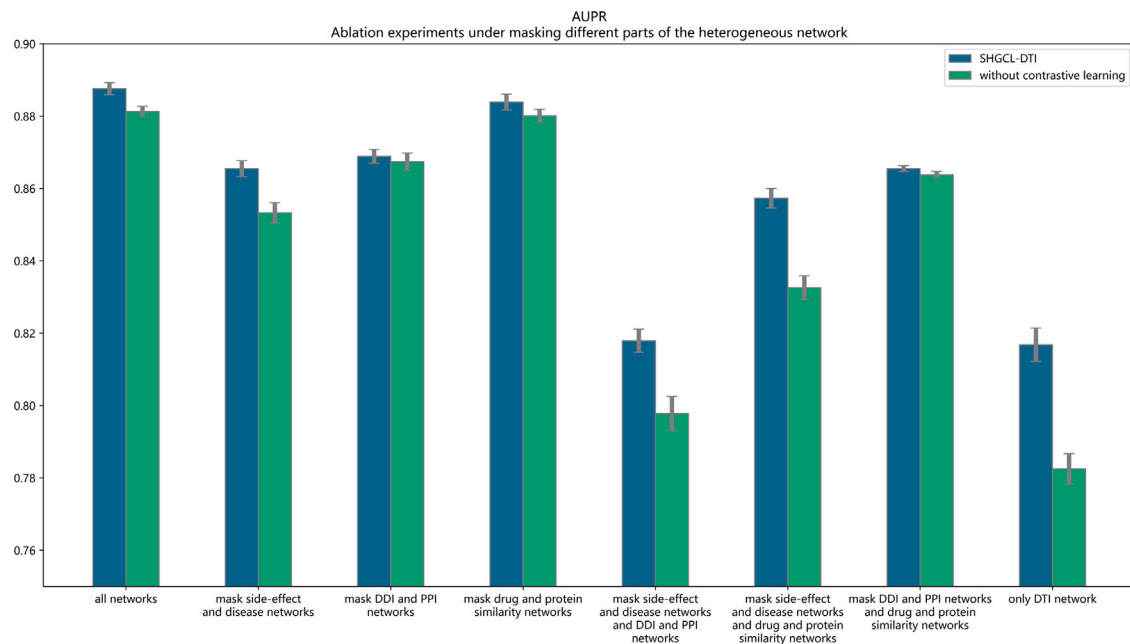


Fig. 5. The AUPR of the results of the ablation experiments in 8 cases. We divide 7 networks except DTI network into 3 groups. Depending on whether to mask them or not, there are 8 cases. Ablation experiments are conducted in the eight cases. DDI network and PPI network represent drug–drug interaction network and protein–protein interaction network, respectively.

a part of the known DTIs. The set of the unmasked DTIs denotes the positive samples in the training set, and the set of masked DTIs denotes the positive samples in the testing set. The negative samples are randomly selected from the unknown DTIs, whose number is 10 times as many as the positive samples. The ratios of randomly masking the known DTIs are {10%, 30%, 50%, 70%, 90%}. As shown in Fig. 4, although the AUROC of SHGCL-DTI is very close to the second best method after masking more than 30% of the known DTIs, SHGCL-DTI outperforms the other baseline methods on AUPR with 3.24%, 1.12%, 1.00%, 0.63% and 1.80% improvements than the second best method, respectively. The experiment results demonstrate that SHGCL-DTI also has the best performance in predicting the unknown DTIs.

It should be noted that EEG-DTI does not perform well on AUPR in scenarios (b) and (f) as shown in Fig. 3. EEG-DTI only learns from positive samples in the training set, which accounts for its poor AUPR in both scenarios. The detailed PR curves can be found in Supplementary Figure S1 and S2. Unlike EEG-DTI, SHGCL-DTI makes apply of the knowledge from negative samples with contrastive learning and achieves the highest AUPR among all compared methods.

3.3.2. Ablation study

In this section, we perform ablation experiments by masking specific edges of the heterogeneous network to demonstrate the improvement of the contrastive learning module in the performance of DTI prediction and generalization ability against the lack of heterogeneous information. We divided the other seven networks except the DTI network into three groups: the DDI and PPI networks, the similarity networks of drug and protein, and disease and side-effect related networks. There are $2^3 = 8$ cases depending on whether to mask the three groups of networks. We performed ablation experiments in these 8 cases. We consider SHGCL-DTI as a complete framework, and design a variant by masking the contrastive learning module. We test the performance of the DTI prediction of them under the condition of keeping the rest of the settings the same.

The results of AUPR and AUROC are shown in Fig. 5 and Supplementary Figure S3, respectively. After masking different parts of the heterogeneous network, it can be observed that the performance of the framework decreases. Through the ablation experiments in eight

cases, it can be found that our complete framework outperforms the one without contrast learning module in both AUPR and AUROC. When only DTI network is left, the improvement on both of the AUROC and AUPR is 1.45% and 3.43%, respectively. This indicates that the contrastive learning module can provide better generalization ability and improve the performance of DTI prediction for our framework. However, the improvement of the contrastive learning module is not significant in the cases of masking the DDI and PPI networks or masking both of the similarity networks and the DDI and PPI networks. This is mainly due to the fact that the DDI network and the PPI network are involved in our meta-path selection. Masking the DDI and PPI networks affects our choices of the meta-paths, which limits the improvement of the prediction performance of the contrastive learning module.

With the above results, we can draw the following two conclusions: (1) Except for the two cases mentioned above which affect the meta-paths, the contrastive learning module can significantly improve the prediction performance. This demonstrates that the contrastive learning module improves the generalization ability and prediction performance of SHGCL-DTI. (2) Integrating more networks can assist the framework in learning the representations of nodes more accurately and can improve the performance of DTI prediction.

3.3.3. Initialization strategy analysis

In this section, we analyze the impact of different initialization methods on the prediction results of DTIs. We classify the initialization strategies adapted in our experiments into the following three types:

Random strategy. The four types of nodes are initialized using a random initialization strategy. In order to have a similar input form to Luo's strategy and SHGCL-DTI's strategy, the input dimension of each node is 128.

Luo's strategy [1]. Luo provides an initialization method to obtain 100-dimensional drug features and 400-dimensional protein features. We directly apply Luo's publicly available initialization results. In addition, since Luo does not provide side effects and disease features, we apply random initialization for these two types of nodes.

Our strategy. SHGCL-DTI, as mentioned above, applies molecular fingerprints based on drug SMILES expressions to obtain drug node features. SHGCL-DTI obtains protein node features based on the relative ratio of amino acids and dipeptides in protein sequences.

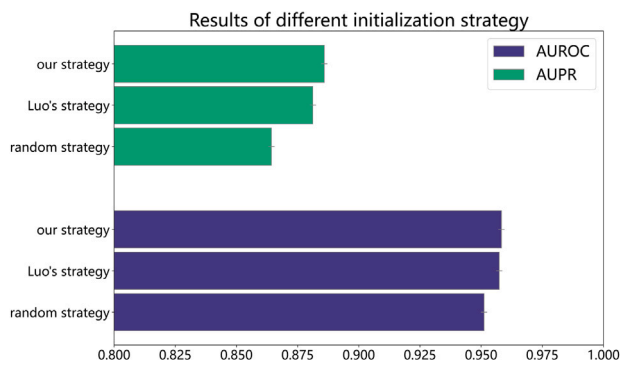


Fig. 6. Results of different initialization strategies.

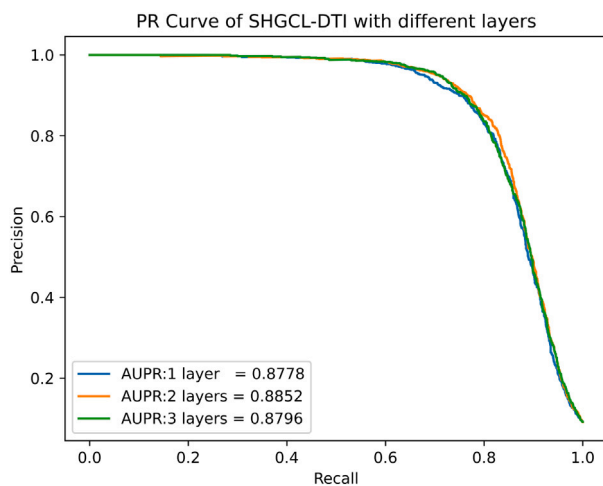


Fig. 7. Precision-Recall Curve of SHGCL-DTI with different layers.

From Fig. 6, we can find that the random initialization strategy gets the worst results, while the best result is obtained by our feature initialization strategy. Compared with Luo's initialization strategy which obtains the second best results, SHGCL-DTI achieves 0.17% and 0.59% improvement in AUROC and AUPR. It is because our strategy can supplement the information of the node itself which is what random strategy and Luo's strategy lacks of. And these results demonstrate that a good initialization strategy can improve the prediction performance of DTIs.

3.3.4. Hyper-parameter sensitivity analysis

We conduct sensitivity analysis for SHGCL-DTI's hyper-parameters to find the best framework structure. We first vary the number of GCN layers l , since a deeper GCN may meet the problem of gradient disappearance and perform worse [47]. We plan to select l from 1 to 5 and draw the difference in Fig. 7 and Supplementary Figure S4. It can be seen from Fig. 7 that the framework obtains the best performance when the number of layers is set to 2. As the training time increases significantly when $l \geq 4$, we do not continue to increase the number of layers.

We also change the dimensionality of the hidden layers to explore their impact on prediction results. We select d from {256, 512, 1024, 2048, 3072}. As shown in Fig. 8 and Supplementary Figure S5, AUPR keeps stable when the dimensionality of the hidden layer is larger than 1024. Considering the time consumption and the prediction performance, we choose 2048 as the dimensionality of the hidden layer.

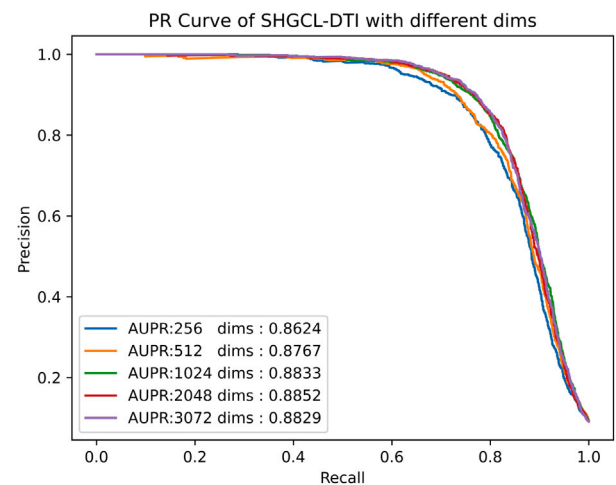


Fig. 8. Precision-Recall Curve of SHGCL-DTI with different dims.

3.3.5. Prediction result verification with literature support

SHGCL-DTI is designed to discover the potential novel DTIs, so we analyze the predicted DTIs with high prediction score (>0.95) in this section. Specifically, we use the whole heterogeneous network to train our model and obtain the prediction result. We exclude the known DTIs from the result and select the top 20 predicted DTIs according to their prediction scores. There are 17 predicted DTIs supported by previous studies among the top 20 predicted DTIs, which means that the valid prediction result accounts for 85% at least. Supplementary Table S4 shows the 17 predicted DTIs. We also try to remove these DTIs which can be easily predicted (drug chemical structure similarities > 0.6 and protein sequence similarities > 0.4). Four easily predicted DTIs are removed and the proportion of the valid prediction result becomes 81.25%.

For example, aripiprazole, an atypical antipsychotic medication used for the treatment of schizophrenia, is predicted to interact with HTR2B which can function as a receptor for various ergot alkaloid derivatives and psychoactive substances.[50] This interaction is reported by previous study [51] that aripiprazole was an inverse agonist at HTR2B receptors, which decreased PI hydrolysis to approximately half of basal levels. What is more, glycopyrrolate is predicted by SHGCL-DTI to interact with CHRM2. Glycopyrrolate is in a class of medications called anticholinergics and CHRM2 (cholinergic receptor muscarinic 2) is involved in mediation of bradycardia and a decrease in cardiac contractility.[52] This prediction can be supported by previous study where Glycopyrrolate is reported to be non-selective muscarinic M_2 (CHRM2) antagonists.[53] The detailed information about the 17 predicted DTIs with biological literature support is shown in Supplementary Table S4. According to the analysis with literature, we find that SHGCL-DTI is able to provide reliable potential DTIs.

4. Conclusion

In this paper, we propose a DTI prediction framework based on heterogeneous graph contrastive learning, called SHGCL-DTI, which mitigates the problem that the known DTIs are very limited and expensive to obtain. We construct a heterogeneous network by combining multiple biological networks. Then we supplement the classical supervised DTI prediction task with an auxiliary graph contrastive learning task which utilizes node representations from the neighbor view and the meta-path view. We conduct comprehensive experiments to simulate a variety of challenging scenarios on public dataset. The results show significant improvements in the cases of unbalanced labels, less redundant DTIs, no priori knowledge and masking partial known DTIs,

by comparing SHGCL-DTI with other baselines. We prove the effectiveness of the contrastive learning module which can improve the generalization ability and prediction performance of SHGCL-DTI. In addition, the impacts of different node initialization strategies, GCN layers, and hidden layer dimensions on the prediction of DTIs are also tested. Finally, according to the prediction result verification with biological literature support, the reliability of the predicted DTIs can be verified.

In summary, our study shows that our contrastive learning module can improve the generalization ability and performance of DTIs prediction. SHGCL-DTI is expected to be a powerful and useful tool to provide potential DTIs for drug discovery and drug repositioning.

CRediT authorship contribution statement

Kainan Yao: Conceptualization, Methodology, Data processing, Validation, Analysis of results, Writing – original draft. **Xiaowen Wang:** Conceptualization, Methodology, Analysis of results, Writing – review & editing. **Wannian Li:** Methodology, Analysis of results, and Writing – review & editing. **Hongming Zhu:** Conceptualization, Methodology, Analysis of results, Writing – review & editing. **Yizhi Jiang:** Methodology, Analysis of results, Writing – review & editing. **Yulong Li:** Data processing, Writing – review & editing. **Tongxuan Tian:** Data processing, Writing – review & editing. **Zhaoyi Yang:** Conceptualization, Methodology, Writing – review & editing. **Qi Liu:** Conceptualization, Methodology, Writing – review & editing. **Qin Liu:** Conceptualization, Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.combiomed.2023.107199>. Supplementary data are available online at *computers in biology and medicine*.

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